



Factoring Quadratic Expressions

Factorising Quadratic Expressions

A quadratic expression is an expression of the form $ax^2 + bx + c$, $a \neq 0$

a is the co-efficient of the x^2 term.

b is the co-efficient of the x term.

c is the constant term.

Some quadratic expressions can be factorised by taking out a factor and using a single bracket, others need a double bracket. This presentation assumes that you are completely familiar with expansion of double brackets by “inspection”, i.e. you can do it mentally.

So for example you
should be able to
expand:

$$(2x + 3)(3x - 4) \rightarrow 6x^2 + x - 12$$

More-or-less
instantly

Common Factor Expressions

A quadratic expression is an expression of the form $\underline{ax^2 + bx + c}$, $a \neq 0$

Taking out a single factor from a **binomial expression**.

Example 1

$$\begin{aligned}\text{Factorise: } & x^2 + 7x \\ & = x(x + 7)\end{aligned}$$

Example 2

$$\begin{aligned}\text{Factorise: } & 12x^2 - 8x \\ & = 4x(3x - 2)\end{aligned}$$

Example 3

$$\begin{aligned}\text{Factorise: } & 9p^2 - 3p \\ & = 3p(3p - 1)\end{aligned}$$

Example 4

$$\begin{aligned}\text{Factorise: } & 8q + 20q^2 \\ & = 4q(5q + 2)\end{aligned}$$

Common Factor Expressions

A quadratic expression is an expression of the form $ax^2 + bx + c$, $a \neq 0$

Taking out a single factor from a **binomial expression**.

Factorise the following:

(a) $x^2 + 8x$
(b) $2x^2 - 4x$
(c) $15x^2 - 10x$
(d) $12x^2 + 18x$
(e) $9y^2 + 3y$
(f) $10k + 15k^2$

Common Factor Expressions

A quadratic expression is an expression of the form $ax^2 + bx + c$, $a \neq 0$

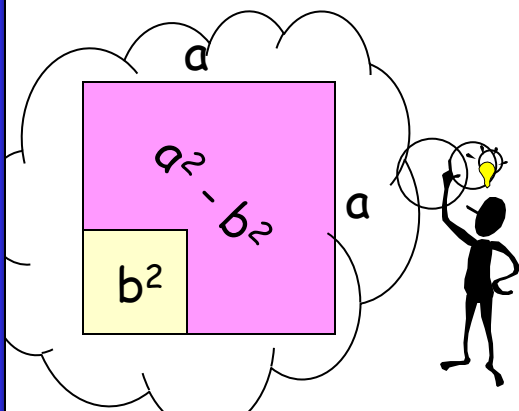
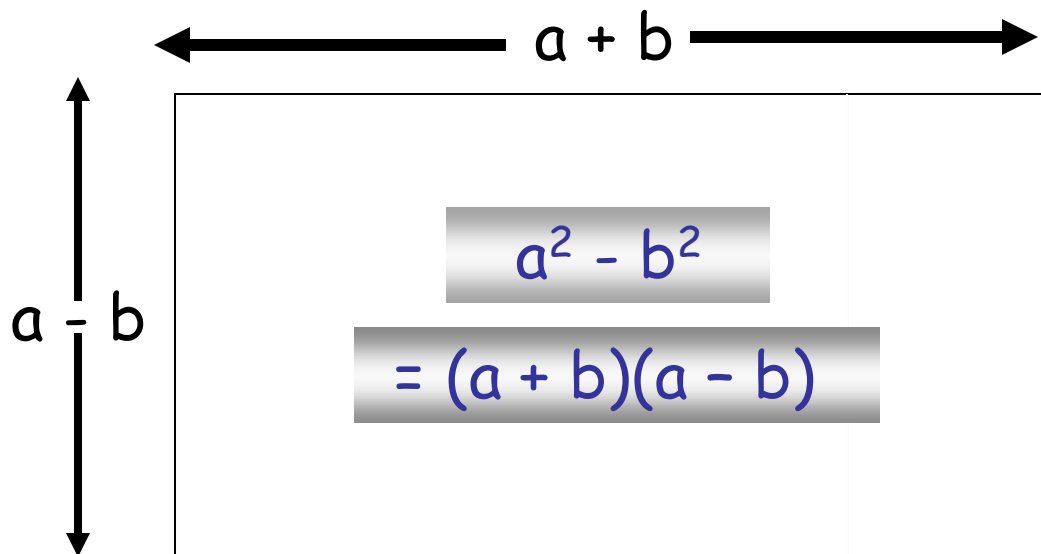
Taking out a single factor from a **binomial expression**.

Factorise the following:

(a) $x^2 + 8x$	$= x(x + 8)$
(b) $2x^2 - 4x$	$= 2x(x - 2)$
(c) $15x^2 - 10x$	$= 5x(3x - 2)$
(d) $12x^2 + 18x$	$= 6x(2x + 3)$
(e) $9y^2 + 3y$	$= 3y(3y + 1)$
(f) $10k + 15k^2$	$= 5k(3k + 2)$

The Difference of Two Squares

To show geometrically that $a^2 - b^2 = (a + b)(a - b)$



< continue with trinomials when ready

Difference of 2 Squares Expressions

A quadratic expression is an expression of the form $ax^2 + bx + c$, $a \neq 0$

Factorising by completing the square from a **binomial expression**.

Quadratic expressions of the form $x^2 - y^2$ are easily factorised by the method of completing the square.

$$a^2 - b^2 = (a + b)(a - b)$$

This result is important as well as being very useful in certain arithmetic calculations that we will look at shortly. It should be committed to memory.

Difference of 2 Squares Expressions

A quadratic expression is an expression of the form $\underline{ax^2 + bx + c}$, $a \neq 0$

Factorising by completing the square from a binomial expression.

Turning to some algebraic expressions now and factorising each in turn.

$$a^2 - b^2 = (a + b)(a - b)$$

Example 1

$$\begin{aligned}x^2 - 16 \\&= x^2 - 4^2 \\&= (x + 4)(x - 4)\end{aligned}$$

Example 2

$$\begin{aligned}y^2 - 1 \\&= (y + 1)(y - 1)\end{aligned}$$

Example 3

Use a single step
once you are
used to these

$$\begin{aligned}9x^2 - 16y^2 \\&= (3x)^2 - (4y)^2 \\&= (3x + 4y)(3x - 4y)\end{aligned}$$

Example 4

$$\begin{aligned}a^2 - 4b^2 \\&= (a + 2b)(a - 2b)\end{aligned}$$

Difference of 2 Squares Expressions

A quadratic expression is an expression of the form $ax^2 + bx + c$, $a \neq 0$

Factorising by completing the square from a **binomial expression**.

Factorise the following:

$$a^2 - b^2 = (a + b)(a - b)$$

(a) $m^2 - n^2$

(b) $x^2 - 25$

(c) $4x^2 - 36$

(d) $25a^2 - 16b^2$

(e) $-1 + 9y^2$

(f) $100k^2 - 9m^2$

Difference of 2 Squares Expressions

A quadratic expression is an expression of the form $\underline{ax^2 + bx + c}$, $a \neq 0$

Factorising by completing the square from a **binomial expression**.

Factorise the following:

$$a^2 - b^2 = (a + b)(a - b)$$

(a) $m^2 - n^2$	$= (m + n)(m - n)$
(b) $x^2 - 25$	$= (x + 5)(x - 5)$
(c) $4x^2 - 36$	$= (2x + 6)(2x - 6)$
(d) $25a^2 - 16b^2$	$= (5a + 4b)(5a - 4b)$
(e) $-1 + 9y^2$	$= (3y + 1)(3y - 1)$
(f) $100k^2 - 9m^2$	$= (10k + 3m)(10k - 3m)$

Factorising Trinomial Expressions

A quadratic expression is an expression of the form **$ax^2 + bx + c$** , $a \neq 0$

Factorising **trinomial expressions**

The simplest quadratic expressions of this type to factorise are those where the co-efficient of x^2 is 1. This can be done using trial and error/improvement and is simply the reverse of expanding double brackets.

Example 1

Factorise: $x^2 + 7x + 12$
 $= (x + 3)(x + 4)$

1. Write the double bracket with the x 's in the usual position.
2. Find 2 numbers whose product is 12 and whose sum is 7.
3. In this simple case there are no complications with signs and the numbers are 3 and 4. Complete the bracket entries. In this case the order does not matter.

Factorising Trinomial Expressions

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Factorising **trinomial expressions**

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Example 1

$$\begin{aligned}\text{Factorise: } x^2 + 7x + 12 \\ = (x + 3)(x + 4)\end{aligned}$$

Example 2

$$\begin{aligned}\text{Factorise: } x^2 + 8x - 20 \\ = (x + 10)(x - 2)\end{aligned}$$

1. Write the double bracket with the x 's in the usual position. One of the signs must be -ve because of the - 20

2. Find 2 numbers whose product is -20 and whose sum is 8.

3. Trying various combinations.

- 4 and 5 ✗,
4 and - 5 ✗,
10 and - 2 ✓,

Factorising Trinomial Expressions

A quadratic expression is an expression of the form $ax^2 + bx + c$, $a \neq 0$

Factorising trinomial expressions

The simplest quadratic expressions of this type to factorise are those where the co-efficient of x^2 is 1. This can be done using trial and error/improvement and is simply the reverse of expanding double brackets.

Example 3

$$\begin{aligned}\text{Factorise: } x^2 - 6x + 8 \\ = (x - 4)(x - 2)\end{aligned}$$

1. Both signs must be negative since we need some negative x as well as a positive constant.

2. Find 2 negative numbers whose product is 8 and whose sum is -6.

3. Trying various combinations.

- 1 and -8 ✗,
-4 and -2 ✓,

Factorising Trinomial Expressions

A quadratic expression is an expression of the form $ax^2 + bx + c$, $a \neq 0$

Factorising trinomial expressions

The simplest quadratic expressions of this type to factorise are those where the co-efficient of x^2 is 1. This can be done using trial and error/improvement and is simply the reverse of expanding double brackets.

Example 3

$$\begin{aligned}\text{Factorise: } x^2 - 6x + 8 \\ = (x - 4)(x - 2)\end{aligned}$$

Example 4

$$\begin{aligned}\text{Factorise: } x^2 + 4x - 12 \\ = (x + 6)(x - 2)\end{aligned}$$

1. Write the double bracket with the x 's in the usual position. One of the signs must be -ve because of the - 12
2. Find 2 numbers whose product is -12 and whose sum is 4.
3. Trying various combinations.
4 and -3 ✗,
6 and -2 ✓,

Factorising Trinomial Expressions

A quadratic expression is an expression of the form $ax^2 + bx + c$, $a \neq 0$

Factorising **trinomial expressions**

Factorise the following:

(a) $x^2 + 3x + 2$

(b) $x^2 + 11x + 10$

(c) $x^2 + 3x - 10$

(d) $x^2 + x - 12$

(e) $x^2 - 6x + 9$

(f) $x^2 - 13x + 12$

(g) $y^2 - 5y - 24$

Factorising Trinomial Expressions

A quadratic expression is an expression of the form $\underline{ax^2 + bx + c}$, $a \neq 0$

Factorising **trinomial expressions**

Factorise the following:

(a) $x^2 + 3x + 2$	$= (x + 1)(x + 2)$
(b) $x^2 + 11x + 10$	$= (x + 10)(x + 1)$
(c) $x^2 + 3x - 10$	$= (x + 5)(x - 2)$
(d) $x^2 + x - 12$	$= (x + 4)(x - 3)$
(e) $x^2 - 6x + 9$	$= (x - 3)(x - 3)$
(f) $x^2 - 13x + 12$	$= (x - 1)(x - 12)$
(g) $y^2 - 5y - 24$	$= (y + 3)(y - 8)$

Factorising Higher Order Trinomial Expressions

A quadratic expression is an expression of the form $\underline{ax^2 + bx + c}$, $a \neq 0$

Factorising **trinomial expressions**

When the coefficient of x is greater than 1, factorising quadratics becomes more tricky when using the previous method. This is particularly the case when a has several factors such as in the number 6. There are more combinations of x entries to start with.

Then (in this case) there are all the factors of 12 together with the signs to consider

$$\begin{aligned} &6x^2 - x - 12 \\ &= (6x + \quad)(x - \quad) \\ &= (x + \quad)(6x - \quad) \\ &= (3x + 4)(2x - 3) \\ &= (2x + \quad)(3x - \quad) \end{aligned}$$

You can still use this method of trial and error/improvement or you may prefer to use a more mechanical method as described next.

1 and - 12

-12 and 1

2 and - 6

6 and -2

3 and - 4

4 and -3 ✓

Factorising Higher Order Trinomial Expressions

A quadratic expression is an expression of the form $\underline{ax^2 + bx + c}$, $a \neq 0$

Factorising **trinomial expressions**

This method is straightforward and relies on **factorising in pairs**.

Example 1

$$6x^2 + 11x + 4$$

$$\underline{6x^2 + 3x} + \underline{8x + 4}$$

$$3x(2x + 1) + 4(2x + 1)$$

$$(2x + 1)(3x + 4)$$

1. Multiply **a** and **c** (6×4) = 24 and find the two numbers whose product is 24 and whose sum is 11.
2. These are 3 and 8.
3. Re-write the expression splitting the **bx** term into two components using these factors.
4. Factorise in pairs taking the HCF of each.
5. The factor in brackets is common to both.

Factorising Higher Order Trinomial Expressions

A quadratic expression is an expression of the form **$ax^2 + bx + c$** , $a \neq 0$

Factorising **trinomial expressions**

This method is straightforward and relies on **factorising in pairs**.

Example 2

$$4x^2 + x - 5$$

$$\underline{4x^2 + 5x} - \underline{4x + 5}$$

$$x(4x + 5) - 1(4x + 5)$$

$$(4x + 5)(x - 1)$$

1. Multiply **a** and **c** (4×-5) = -20 and find the two numbers whose product is -20 and whose sum is 1.
2. These are 5 and -4.
3. Re-write the expression splitting the **bx** term into two components using these factors.
4. Factorise in pairs taking the HCF of each.
5. The factor in brackets is common to both.

Factorising Higher Order Trinomial Expressions

A quadratic expression is an expression of the form $\underline{ax^2 + bx + c}$, $a \neq 0$

Factorising **trinomial expressions**

This method is straightforward and relies on **factorising in pairs**.

Example 3

$$5x^2 - 44x + 32$$

$$\underline{5x^2 - 4x} - \underline{40x + 32}$$

$$x(5x - 4) - 8(5x - 4)$$

$$(5x - 4)(x - 8)$$

1. Multiply **a** and **c** (5×32) = 160 and find the two numbers whose product is 160 and whose sum is -44.
2. These are -4 and -40.
3. Re-write the expression splitting the **bx** term into two components using these factors.
4. Factorise in pairs taking the HCF of each.
5. The factor in brackets is common to both.

Factorising Higher Order Trinomial Expressions

A quadratic expression is an expression of the form $\underline{ax^2 + bx + c}$, $a \neq 0$

Factorising **trinomial expressions**

This method is straightforward and relies on **factorising in pairs**.

Finally, tackling the original problem that we looked at:

Example 4

$$6x^2 - x - 12$$

$$\underline{6x^2 - 9x} + \underline{8x - 12}$$

$$3x(2x - 3) + 4(2x - 3)$$

$$(2x - 3)(3x + 4)$$

1. Multiply **a** and **c** (6×-12) = -72 and find the two numbers whose product is -72 and whose sum is -1.
2. These are -9 and 8.
3. Re-write the expression splitting the **bx** term into two components using these factors.
4. Factorise in pairs taking the HCF of each.
5. The factor in brackets is common to both.

Factorising Higher Order Trinomial Expressions

A quadratic expression is an expression of the form $ax^2 + bx + c$, $a \neq 0$

Factorising **trinomial expressions**

Factorise the following:

(a) $8x^2 + 10x + 3$
(b) $6x^2 + 31x + 40$
(c) $8x^2 - 10x - 3$
(d) $5x^2 - 16x + 3$
(e) $9x^2 + 6x - 8$
(f) $12x^2 + x - 1$
(g) $24y^2 - 55y - 24$

Factorising Higher Order Trinomial Expressions

A quadratic expression is an expression of the form $\mathbf{ax^2 + bx + c}$, $a \neq 0$

Factorising **trinomial expressions**

Factorise the following:

(a) $8x^2 + 10x + 3$	$= (2x + 1)(4x + 3)$
(b) $6x^2 + 31x + 40$	$= (3x + 8)(2x + 5)$
(c) $8x^2 - 10x - 3$	$= (4x + 1)(2x - 3)$
(d) $5x^2 - 16x + 3$	$= (5x - 1)(x - 3)$
(e) $9x^2 + 6x - 8$	$= (3x + 4)(3x - 2)$
(f) $12x^2 + x - 1$	$= (3x + 1)(4x - 1)$
(g) $24y^2 - 55y - 24$	$= (8y + 3)(3y - 8)$